How-To Library

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Using Test Data Manager to Generate Test Data

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Abstract

With Test Data Management (TDM), you can generate realistic test data for test environments. Import target metadata into a project to define the type of data that you want to generate. This article describes how to create data generation rules, create an entity, and create and run a data generation plan.

Supported Versions

- Test Data Management 9.6.0

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Overview

Use Test Data Manager to generate realistic test data for test environments.

You create standard data generation rules that use predefined generation techniques. Define ad hoc data generation rules within a project for a column in the target table. Create a data generation plan. Run the workflow to generate the test data.

Scenario

You work for an organization that sells plane tickets. You want to generate data in a table that contains customer information such as names and customer identification numbers. You might need an additional table to store ticket numbers and issue dates.

To generate the test data, perform the following tasks:

1. Create a project and import data sources.
2. Create standard data generation rules.
3. Add standard data generation rules to the project.
4. Assign standard data generation rules to the target columns.
5. Create and assign an ad hoc generation rule to the target column.
6. Create an entity to establish the relationship between two tables.
7. Create a data generation plan.
8. Generate, run, and monitor the workflow.

Prerequisites

Before you begin, you must create tables and relationships in the target database. You must install and configure TDM.

In Test Data Manager, perform the following tasks:

- Create a TDM connection to a target database where you want to generate data.
- Add a flat file dictionary that contains first and last names.

Step 1. Create a Project and Import Data Sources

In Test Data Manager, create a project folder to store the data generation components. Import target metadata to the project. In this example, the target database is AutoTarget and the target tables are CUSTOMERS and TICKET tables.

1. In Test Data Manager, click **Projects** to access the projects.

   A list of projects appears.
2. Click **Actions** > **New**.
3. In the **Create Project** dialog box, enter the following project properties:

   Name

   Enter a name.

   Description

   Enter a description.

   PowerCenter Repository

   The default is the name of the PowerCenter repository.

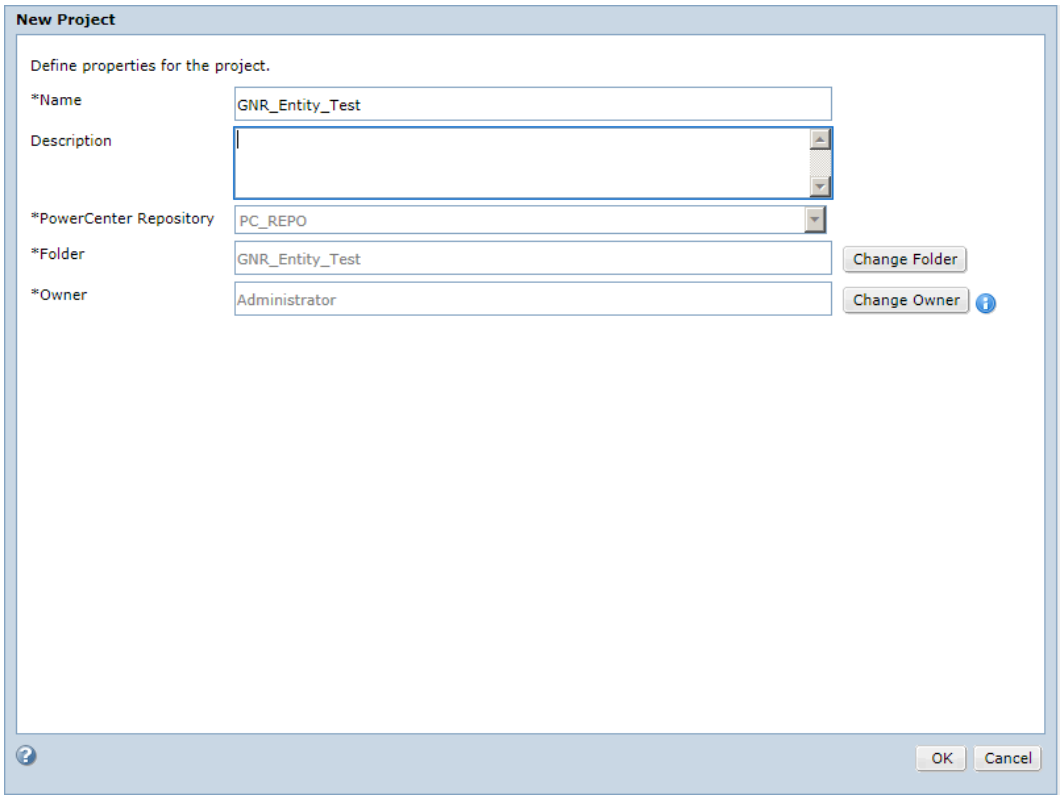
   Folder

   The default is the project name.

   Owner

   The default is the name of the user that created the folder.

The following image shows the properties of the project:



4. Click **OK**.

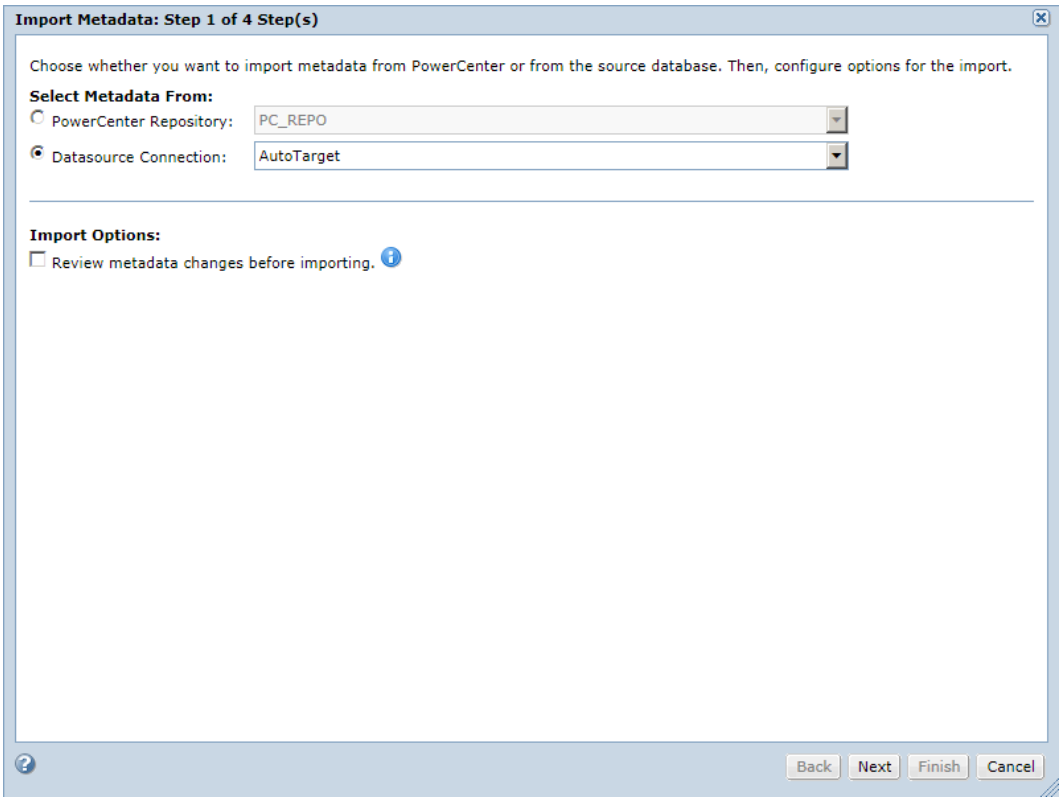
   View the properties of the project that you created.

5. Click **Actions** > **Import Metadata**.

   The **Import Metadata** dialog box appears.

6. To import metadata from a database connection, choose **Datasource Connection** and select AutoTarget.

The following image shows the options to import metadata:



7. Choose whether you want to review the metadata changes before you import the data sources. You can choose to skip the import options.
8. Click **Next**.
9. Select the metadata schema that you want to import from the database connection. You can filter folders by the folder name or the description.
10. Click **Next**.
11. Select all the tables to import.
12. Click **Next**.
13. To import the data source immediately, select **Import Now**.
14. Click **Finish**.

    While the import is running, view the progress of the import job in the **Monitor** view. After the import job finishes, you can access the imported metadata through the **Data Sources** details view.

Step 2. Create Standard Data Generation Rules

You want to generate customer names, identification numbers, and ticket numbers in the target databases. Use standard generation techniques to create dictionary, numeric sequence, and numeric random rules.

To generate data into target columns, perform the following tasks:

- Create a dictionary generation rule to load customer names from a dictionary.
- Create a numeric sequence generation rule to generate identification numbers.
- Create a numeric random generation rule to generate ticket numbers.

Creating a Dictionary Generation Rule

Create a dictionary generation rule to load names from a dictionary into the target table. Select a column from the dictionary to load names in the CUSTOMERS table.

1. In Test Data Manager, click **Policies**.
2. Click **Actions** > **New** > **Generation Rule**.

   The **Rule Wizard** appears.
3. Enter a name and optional description for the rule.
4. Select the **String** data type.
5. Click the **Standard Rule** option and select **Dictionary**.
6. To override generation parameters for a rule, select the **Override Allowed** option.
7. Click **Next**.
8. Enter the following parameters:

   Dictionary Name

   Click **Select** and choose the dictionary with names.

   Dictionary Column

   Select a dictionary column that contains names.

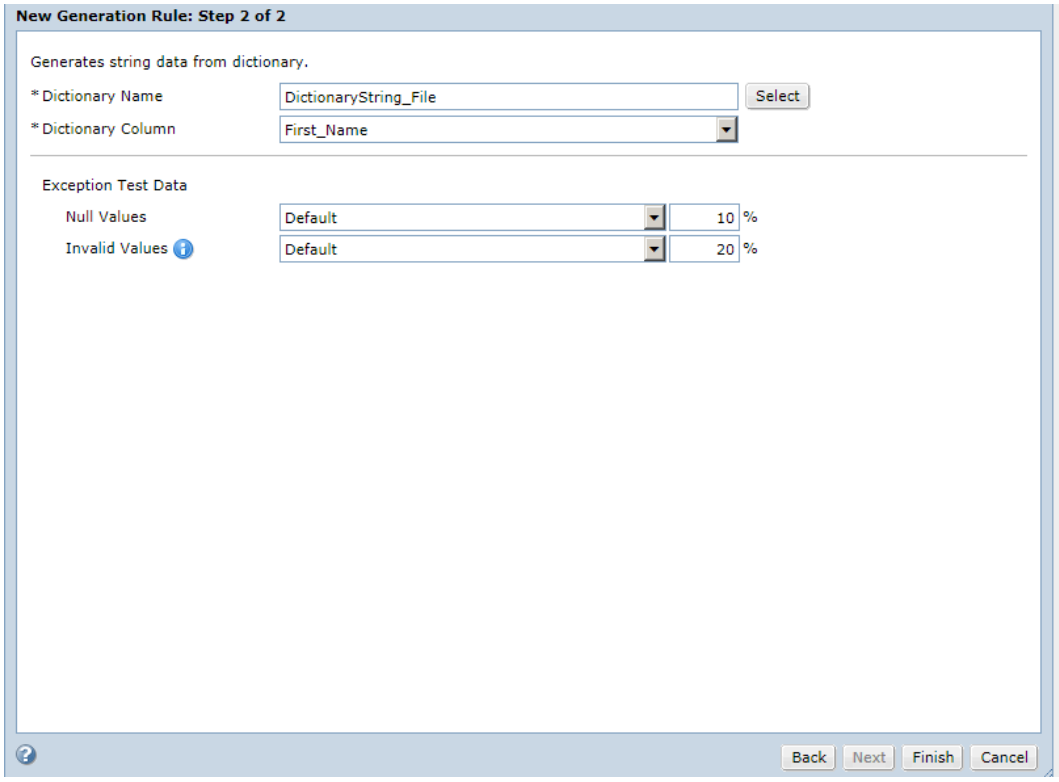
   Null Values

   Enter 10%.

   Invalid Values

   Enter 20%.

   The following image shows the dictionary generation rule parameters:



9. Click **Finish**.

   The rule appears in the list of generation rules.

Creating a Numeric Sequence Generation Rule

Create a numeric sequence generation rule to generate customer identification numbers in the CUSTOMERS table.

1. Click **Policies**.
2. Click **Actions** > **New** > **Generation Rule**.

   The **Rule Wizard** appears.
3. Enter a name and optional description for the rule.
4. Select the **Numeric** data type.
5. Click the **Standard Rule** option and select **Sequence**.
6. To override generation parameters for a rule, select the **Override Allowed** option.
7. Click **Next**.
8. Enter the following parameters:

   Start Value

   Enter 100.

   Increment Value

   Enter 1.

   Null Values

   Enter 10%.

   Invalid Values

   Enter 20%.
9. Click **Finish**.

   The rule appears in the list of generation rules.

Creating a Numeric Random Generation Rule

Create a numeric random generation rule to generate ticket numbers in the TICKET table.

1. Click **Policies**.
2. Click **Actions** > **New** > **Generation Rule**.

   The **Rule Wizard** appears.
3. Enter a name and optional description for the rule.
4. Select the **Numeric** data type.
5. Click the **Standard Rule** option and select **Random**.
6. To override generation parameters for a rule, select the **Override Allowed** option.
7. Click **Next**.
8. Enter the following parameters:

   Minimum Value

   Enter 100.

Maximum Value

Enter 1000.

Generate Unique Values

Select the check box.

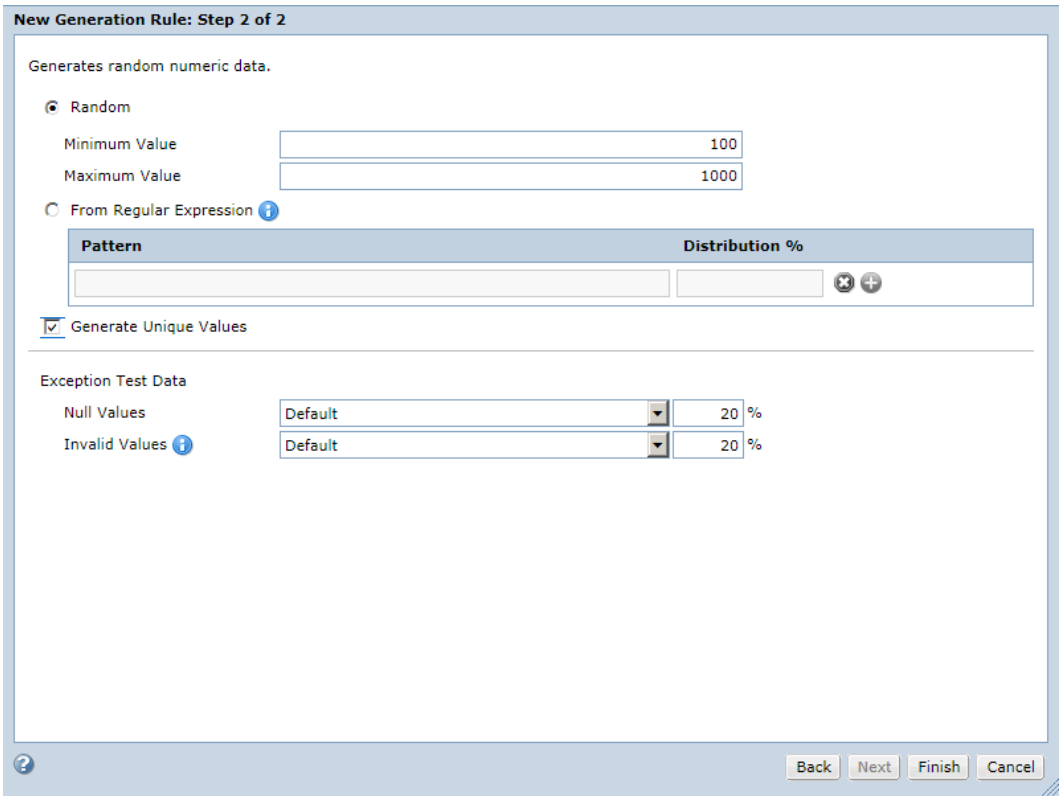
Null Values

Enter 20%.

Invalid Values

Enter 20%.

The following image shows the numeric random generation rule parameters:



9. Click **Finish**.

   The rule appears in the list of generation rules.

Step 3. Add Standard Data Generation Rules to the Project

Add the standard generation rules to the project you created.

1. Click **Projects**.

   You can see a list of projects.

2. Open the project that you created.

   The project window opens in another tab.

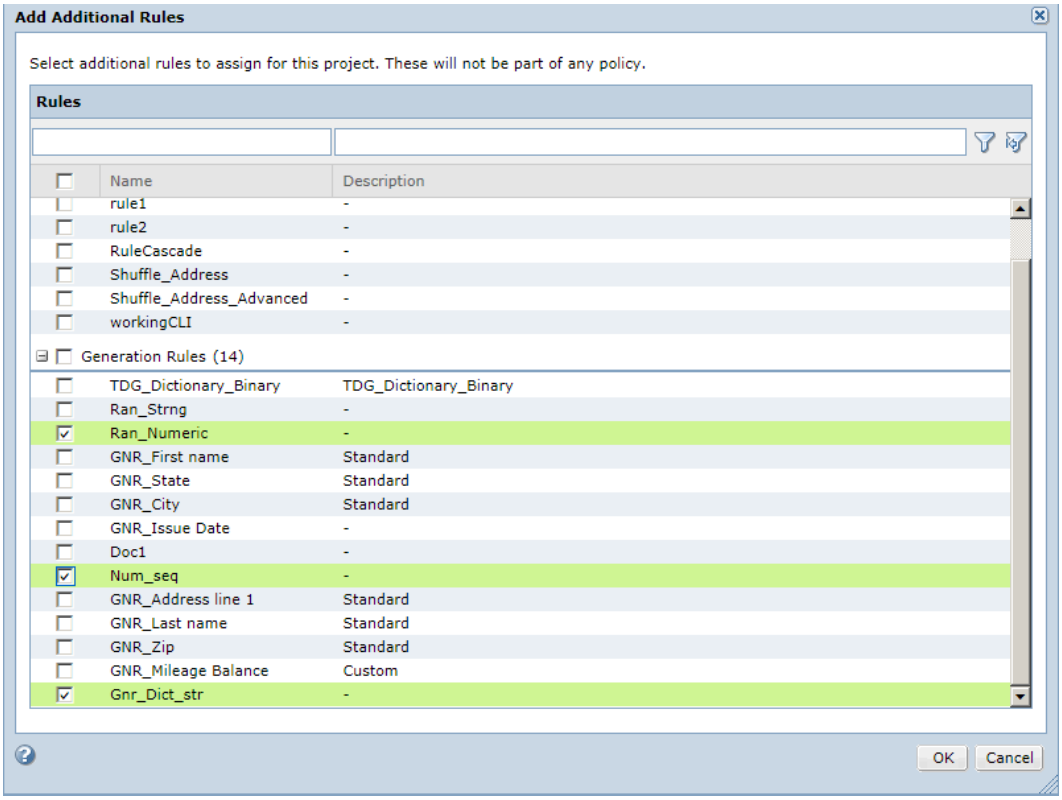
3. Click **Overview** > **Policies**.

4. Click **Actions** > **Add Additional Rules**.

The **Add Additional Rules** dialog box appears.

5. Select the dictionary, random, and sequence generation rules from the list.

   The following image shows a sample list of rules:



6. Click **OK**.

   The rules appear under the **Additional Rules** list.

Step 4. Assign Standard Data Generation Rules

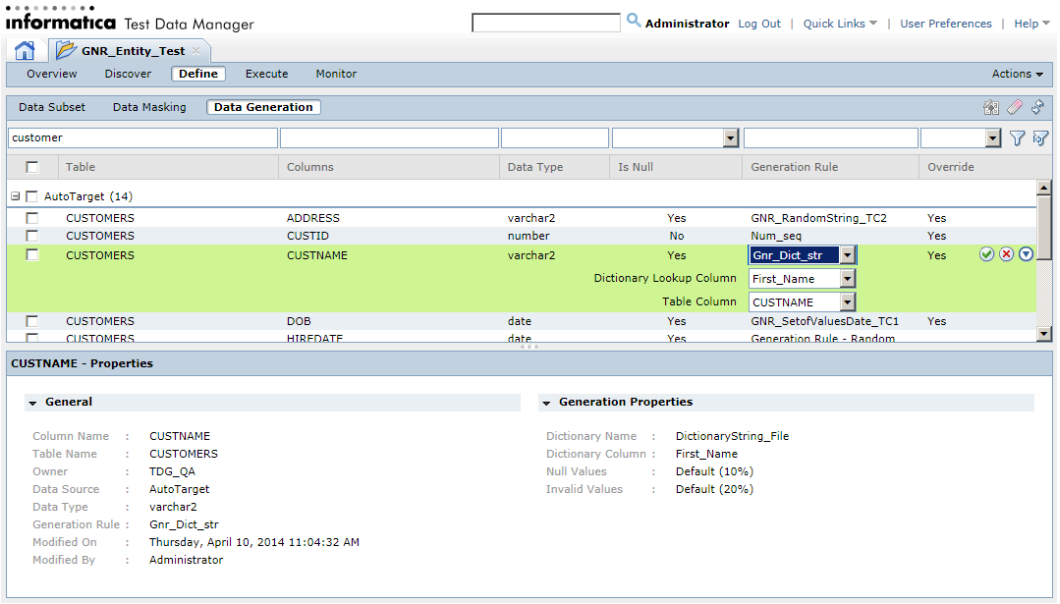
Assign the standard data generation rules to the target columns to generate customer names, customer identification numbers, and ticket numbers. In this example, the target columns are CUSTNAME and CUSTID in the CUSTOMERS table and TICKET_NUMBER in the TICKET table.

1. In the project, click **Define** > **Data Generation**.
2. In the CUSTOMERS table, select the CUSTNAME column.
3. Click inside the **Generation Rule** column.

   A list of data generation rules available for the string data type appears.

4. Select the name of the dictionary generation rule that you created for names.
5. In the Dictionary Lookup Column field, select First_Name.
6. In the Table Column field, select CUSTNAME.

The following image shows a sample dictionary generation rule assignment:



7. To save the rule assignment, click **Save**.
8. In the CUSTOMERS table, select the CUSTID column.
9. Click inside the **Generation Rule** column.
10. Select the numeric sequence generation rule that you created.
11. To save the rule assignment, click **Save**.
12. In the TICKET table, select the TICKET_NUMBER column.
13. Click inside the **Generation Rule** column.
14. Select the numeric random generation rule that you created.
15. To save the rule assignment, click **Save**.

Step 5. Create a Reference Lookup Generation Rule

A reference lookup generation rule is an ad hoc generation rule that you create within the project. Create and assign the reference lookup generation rule to generate issue dates in the ISSUE_DATE column of the TICKET table. Select CUSTOMERS table as the parent table.

1. In the project, click **Define** > **Data Generation** to access the **Data Generation** view.
2. In the TICKET table, select the ISSUE_DATE column to assign the ad hoc generation rule to.
3. Click inside the **Generation Rule** column.

   A list of generation rules that are available for the date data type appears.
4. Click **New Generation Rule**.

   The **New Generation Rule** window appears.
5. Select the **Reference Lookup** generation technique.
6. Enter the following generation parameters:

Minimum Value

Enter 5.

Maximum Value

Enter 50.

Data Source

Select AutoTarget.

Parent Table

Select CUSTOMERS.

Child Column

The default is ISSUE_DATE, the column name on which you create the rule. You cannot remove the first child column.

Parent Column

Select HIREDATE.

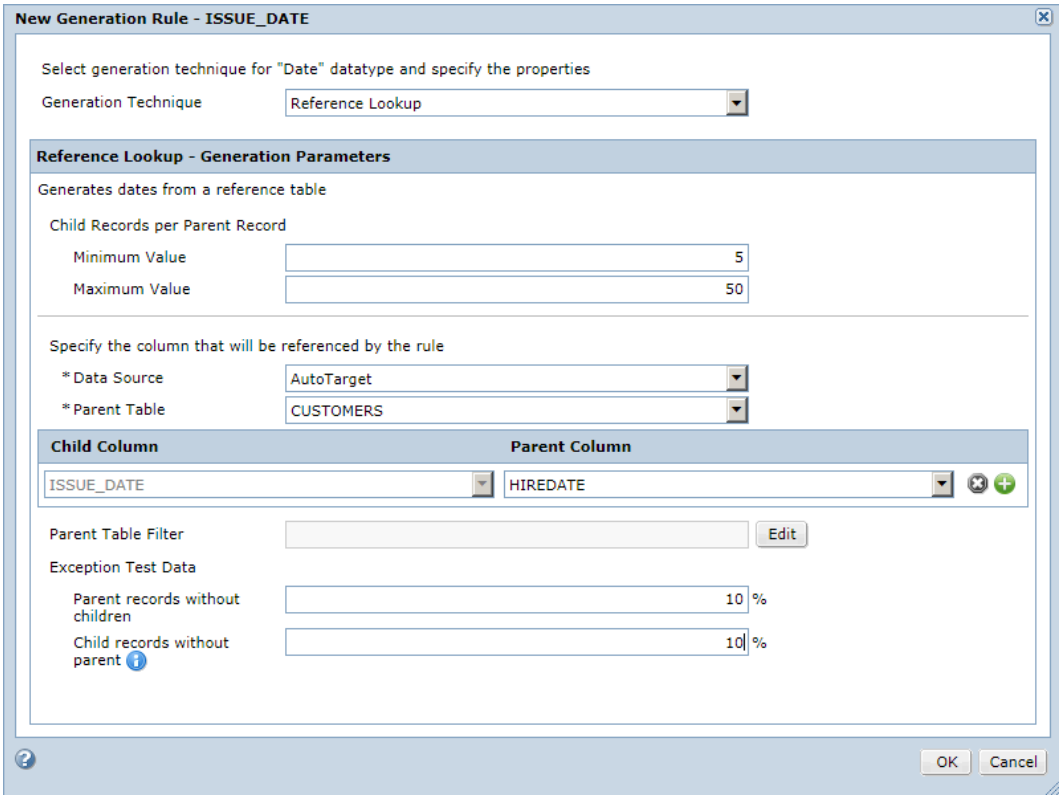
Parent records without children

Enter 10%.

Child records without parent

Enter 10%.

The following image shows the reference lookup generation parameters:



7. To assign the rule, click **OK**.

Step 6. Create an Entity

Create an entity in Test Data Manager to retrieve all tables related to the driving table based on the constraints. Before you create an entity, add foreign key constraints to establish a logical relationship between CUSTOMERS and TICKET tables.

Adding a Foreign Key Constraint

Consider the CUSTOMERS table as parent table and the TICKET table as child table. Establish a logical relationship between the tables with CUSTID as the foreign key.

1. In the project, click **Discover** > **Tables**.
2. Click on the TICKET table.
3. Click **Constraints**.
4. Click **Create New Constraint**.
5. Enter the following constraint properties:

   Name

   Enter CUST_ID.

   Constraint Type

   Select **Foreign Key**.

   Severity Level

   Select **Major**.

   Parent Table

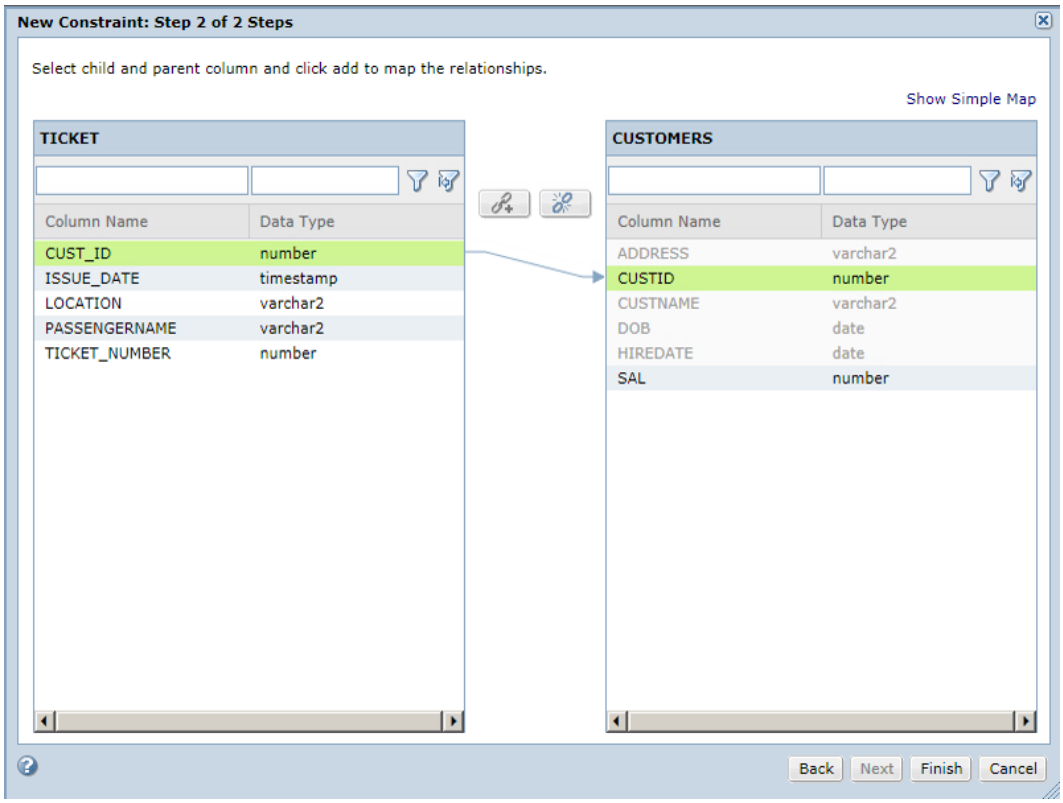
   Click **Select** and choose CUSTOMERS.

   Enable Constraint

   Select the check box.
6. Click **Next**.

   The left pane contains the child table columns and the right pane contains the parent table columns.
7. Click the CUST_ID column from the left pane. Click the CUSTID column from the right pane. Click the **Link** icon to map the parent-child relationship.

The following table shows mapping between CUSTOMERS and TICKET tables:



8. Click **Finish**.

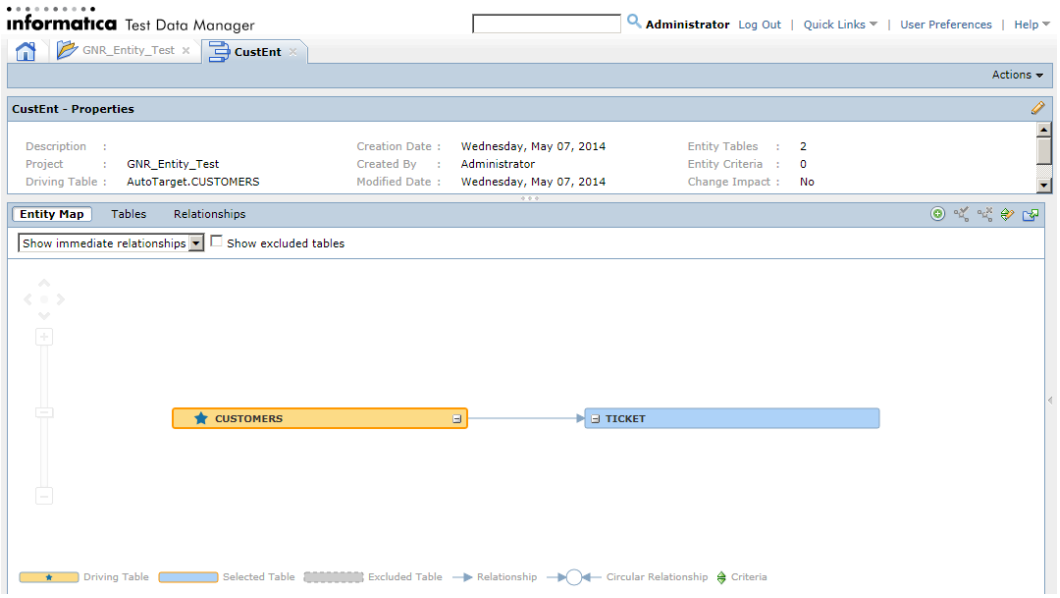
Creating an Entity

Create an entity between CUSTOMERS and TICKET tables based on the foreign key constraint in the CUSTOMERS table.

1. In the project, click **Define** > **Data Subset**.
2. Click **Actions** > **New** > **Entities**.
3. In the **New Entity** dialog box, enter a name and optional description for the entity.
4. To select the driving table, click **Select Table** and choose CUSTOMERS.

5. Click **Save**.

   The following image shows the relationship between CUSTOMERS and TICKET tables in the entity:



Step 7. Create a Data Generation Plan

To perform the data generation operation, you must create a plan. The plan includes data generation components that you need to generate a workflow.

1. In the project, click **Execute**.
2. Click **New Plan**.
3. In the **New Plan** dialog box, enter a name and optional description for the plan.
4. Click **Next**.
5. To skip adding the masking components, click **Next**.
6. To skip adding the subset components, click **Next**.
7. To add data generation operation to the plan, click **Add Generation Components**.
8. Select the entity you created and the CUSTOMERS and TICKET tables to add to the plan.
9. Click **OK**.

   The entity and tables appear in the list.
10. Click **Next**.
11. To specify generation criteria for the entity, select the entity.

    Enter the following values:

    Minimum

    Enter 10.

    Maximum

    Enter 100.
12. Click **Next**.
13. Configure the following plan settings:

Number of Records to Generate

Enter 1000.

Target Connection

Select **Relational**. Click **Select** and choose AutoTarget.

Dictionary Lookup Connection

Select **Flat File**. The default location is populated.

Reference Lookup Connection

Select **Relational**. Click **Select** and choose AutoTarget.

Truncate Tables

Select the check box.

Disable Indexes

Select the check box.

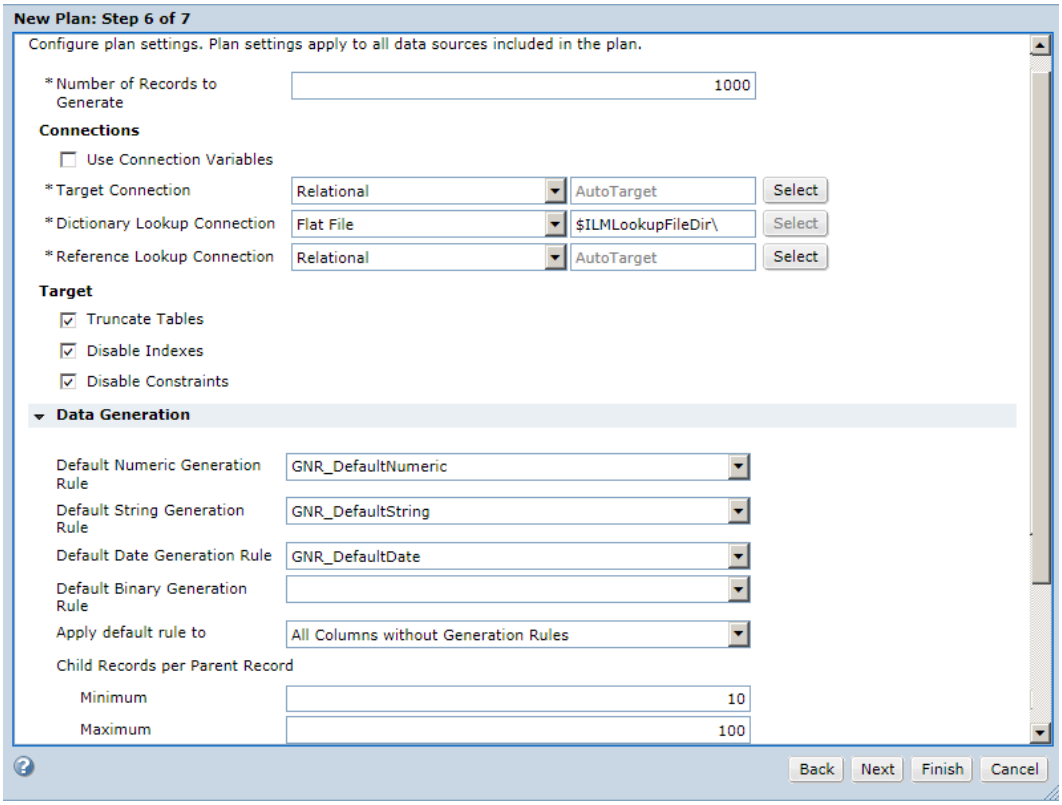
Disable Constraints

Select the check box.

Data Generation

Apply the default rules to the columns that do not have generation rule assignments.

The following image shows a sample data generation plan settings:



14. Click **Next**.
15. Review the data source settings.
16. Click **Finish**.

Step 8. Manage the Workflow

To generate data in the CUSTOMERS and TICKET tables, you must generate and run the workflow. You can monitor the progress of the workflow tasks.

Generating the Workflow

After you create the plan, generate the workflow.

1. In the project, click **Execute** to access the plans in the project.
2. Select the data generation plan you created.
3. Click **Actions** > **Generate Workflow**.

   The **Generate Workflow** dialog box appears.
4. Select **Schedule Now**.
5. Click **Generate Workflow**.

   View the status of the workflow generation in the **Monitor** tab.

Running the Workflow

After you generate the workflow, you must run the workflow to run sessions from the PowerCenter mappings.

1. In the project, click **Execute** to access the plans in the project.
2. Click the data generation plan you created.

   The plan opens in a separate tab.
3. Click **Actions** > **Execute Workflow**.

   The **Execute Workflow** dialog box appears.
4. Select the PowerCenter Integration Service.
5. Select **Schedule Now**.
6. Click **Execute Workflow**.

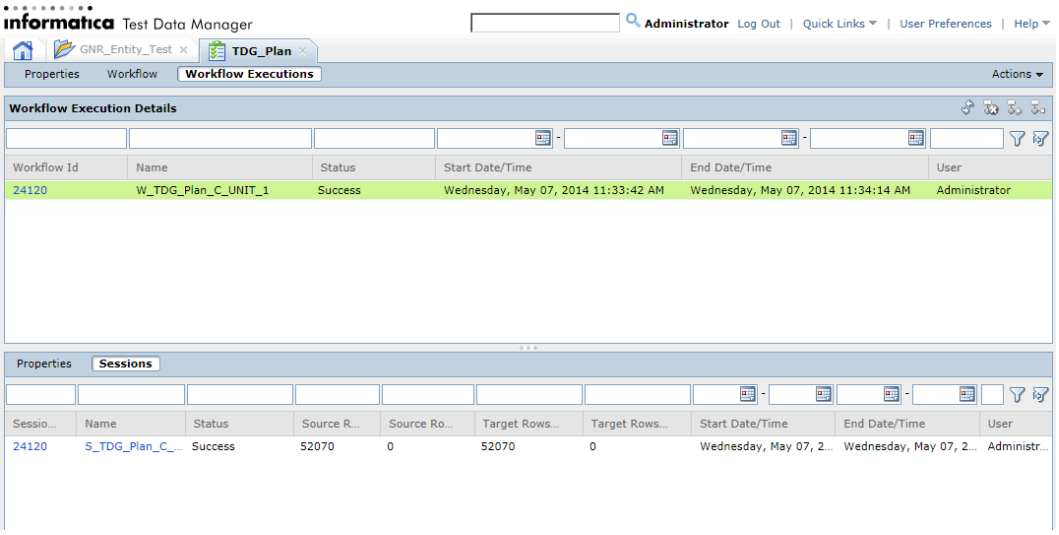
Monitoring the Workflow

You can view the status of the workflow that runs.

1. In the plan, click **Workflow Executions**.
2. To refresh the view, click **Auto Refresh On**.

   View the progress in the Status column.
3. To view the workflow logs, click the Workflow Id.
4. Select the workflow.
5. To view the summary, click the **Properties** tab.
6. To view the session details, click the **Sessions** tab.

The following image shows the workflow details:



7. To view the session logs, click the Session Id.

Author

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